5.1 Conditioning: changing the denominator

Definition and results

For $P(A) > 0$, the probability of B given A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table of equally likely individuals, it is the ratio of the count of $A \cap B$ to the count of A. In general $P_A(B) \neq P_B(A)$.

Status and justification. Definition. Multiplying by $P(A)$, we get the rule $P(A \cap B) = P(A)P_A(B)$. The condition $P(A) > 0$ is essential.

Worked example. Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} =$ 0,3, while $P_B(A) = \dfrac{12}{30} = 0{,}4$. $P(A \cap B) = 0{,}12$.

Watch out

"Among the A" sets the denominator. A conditional probability is not an intersection.

Immediate application. Out of 60 objects, 20 are red; of the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

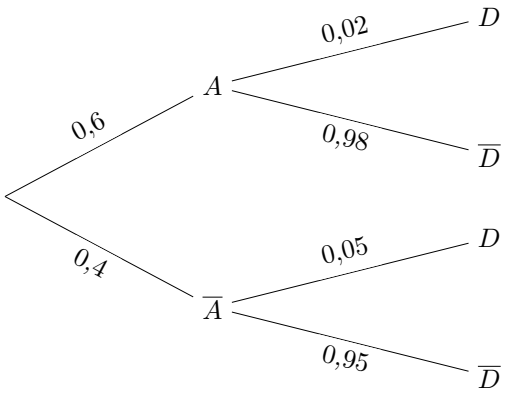
5.2 Trees, partitions and total probability

Definition and results

A partition $A_1, \ldots, A_k$ is a family of disjoint events whose union is the sample space. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. On a tree, the probabilities on branches from the same node sum to 1; probabilities along a path are multiplied; probabilities of disjoint paths are added.

Status and justification. B is the disjoint union of the $B \cap A_i$. By additivity and by the definition of conditional probabilities, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

Worked example. Two machines supply 60% and 40% of the output; their defect rates are 2% and 5%. $P(D) = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05 = 0{,}032$. Among defective items, the first machine represents $\dfrac{0{,}012}{0{,}032} = \dfrac{3}{8}$.



A denotes the machine producing 60 per cent of the output and D a defect. For example, $0{,}02 = P_A(D)$: the second-level probabilities are conditional.

Watch out

The simple mean of the rates is correct only if the branch weights are equal.

Immediate application. $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

5.3 Independence and disjointness

Definition and results

A and B are independent if $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. They are mutually exclusive if $A \cap B = \varnothing$. Two mutually exclusive events with strictly positive probabilities are not independent.

Status and justification. The conditional criterion follows from dividing by $P(A) > 0$. For mutually exclusive events with positive probabilities, the left-hand side is 0 and the product on the right is strictly positive.

Worked example. With a fair die, A: even result, B: result at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: independent. A and "odd result" are mutually exclusive but not independent.

Watch out

Independence in a model must be justified or stated explicitly; "two different events" won't do.

Immediate application. We are given $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

5.4 Successive trials and Bernoulli

Definition and results

A Bernoulli trial has two outcomes: success, with probability p, and failure, with probability 1 $-p$. In a repetition of n identical independent trials, with $n \leq 4$ here, the probabilities of the branches remain p and 1 $-\,p$. A path with k successes has probability $p^k(1-p)^{n-k}$; we counts the paths that realise the event, without the general theory of the binomial distribution.

Status and justification. The probability of a path is a product, by independence. The paths corresponding to an event are disjoint: their probabilities add up.

Worked example. For three trials with $p = \frac{1}{2}$, exactly two successes correspond to SSE, SES, ESS. Each path has probability $\frac{1}{8}$, so the total is $\frac{3}{8}$. At least one success has probability $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

Watch out

When drawing from an urn without replacement, the probabilities generally change after the first draw: the independence model may be wrong.

Immediate application. Four independent trials with success probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

5.5 Simulating and discussing the model

Definition and results

A pseudo-random simulation produces frequencies that fluctuate around a probability of the model. For a uniformly distributed point in the square [0; 1] ^{2}, the proportion satisfying $y \leq x^2$ estimates the area under the parabola; the proportion satisfying $x^2 + y^2 \leq 1$, multiplied by 4, estimates π. The repetitions must use new draws.

Status and justification. Simulation principle assumed and tested. The probability-area equality is an interpretation of the uniform draw; integration is not a prerequisite. A simulation does not prove an exact value.

Worked example. Out of 10000 points, if 7840 are in the quarter disc, the estimate of π is $\frac{4 \times 7840}{10000} = 3{,}136$. Two samples of the same size generally give different values.

Watch out

Fixing a seed makes it possible to reproduce a run, not to remove the randomness of the model or to guarantee a given precision.

Immediate application. A simulation of 2000 points gives 675 points under $y = x^2$. Which area does it estimate?